

EXECUTIVE SUMMARY

Subject-Invariant Spatiotemporal Contrastive Network for Micro-Expression Recognition

Author: Addhyan | Hardware: NVIDIA RTX 4060 | Seed: 42 | Date: May 26, 2026

1. Research Background & Core Challenges

Micro-expressions are transient, involuntary facial muscle contractions lasting less than 500 milliseconds. They serve as standard windows into genuine emotional states but automated Micro-Expression Recognition (MER) using deep learning remains a challenging task. Specifically, traditional deep networks are severely bottlenecked by two twin issues:

- 1. Data Starvation:** Annotation requires certified FACS practitioners to index high-speed clips frame-by-frame. This expensive coding limits benchmark databases (e.g., CASME II has only 255 samples), causing deep architectures to easily overfit or fail during feature extraction.
- 2. Subject Identity Bias:** Static facial details (bone structure, skin texture, lighting) are numerically stronger than dynamic muscle contractions. Deep networks naturally exploit these static identity features. Consequently, models evaluate poorly on unseen subjects, memorizing who the person is rather than what emotion they express.

2. Proposed Four-Stage Pipeline (Methodology)

To address these challenges, we propose a unified, parameter-efficient (~382K parameters) pipeline:

Stage 1: Preprocessing & Motion Amplification

We merge CASME II (200 FPS, pre-cropped) and CAS(ME)² (30 FPS, full frame) into a unified 4-class taxonomy (Negative, Positive, Surprise, Others) to stabilize optimization. To amplify motion, we apply Eulerian Video Magnification (EVM) with a temporal bandpass filter set to 0.4–3.0 Hz (targeting facial muscle contractions) and an amplification factor $\alpha=20$. To strip out static identity details, we extract physics-based optical flow modalities. For each pair of frames, dense Farneback flow is calculated to obtain horizontal flow (u), vertical flow (v), and skin deformation optical strain (os):

$$os = \sqrt{(\partial u / \partial x)^2 + (\partial v / \partial y)^2 + 0.5 * (\partial u / \partial y + \partial v / \partial x)^2}$$

Optical strain magnitude modeling skin shear and compression. Normalizing each channel to [0,1].

Finally, the frames are temporally standardized to exactly L=33 frames (T=32 frame pairs for flow) using linear interpolation.

Stage 2: Hybrid Network Architecture

The network backbone is designed to be highly parameter-efficient to prevent overfitting:

- **Modality-Aware 3D Backbone (STSTNet-3D):** Splitting the 3-channel flow tensor, we route flow-u, flow-v, and strain through three parallel, unshared convolution branches. Weights are unshared to let kernels specialize in each distinct physical motion type. Standard 3D convolutions use a temporal kernel depth of 1, preventing premature temporal mixing.
- **3D SimAM Attention:** To highlight active facial regions (eyebrows, mouth corners) without adding learnable parameters, we integrate 3D SimAM attention. It evaluates analytical energy scores for each neuron, rescaled element-wise based on inter-neuron suppression.
- **SLSTT Sequence Transformer:** Spatial pooling compresses features to shape [B, 32, 96]. We inject sinusoidal positional encodings and process the sequence with a 2-layer pre-norm transformer encoder to learn temporal dynamics. Temporal mean pooling then aggregates the sequence into a 96-d global feature vector.

Stage 3: Joint Domain-Adversarial & Contrastive Learning

The global feature vector is routed through a triple-head wrapper to enforce subject-invariance:

- **Emotion Head:** Classification optimized via class-weighted Focal Loss (gamma=2.0, label smoothing=0.05) to mitigate the severe database class imbalance.
- **Adversarial Identity Head:** Places a Gradient Reversal Layer (GRL) before an identity classification head. During backpropagation, the GRL reverses identity gradients with a coefficient lambda annealed from 0.0 to 1.0, actively stripping identity textures from the spatiotemporal backbone.
- **Contrastive Projection Head:** Maps features to a 64-d space optimized via Supervised Contrastive Loss (SupCon). To bypass batch-size VRAM constraints (B=2), we integrate a Cross-Batch Memory (XBM) queue of size M=64 to cache historical embeddings.

Stage 4: Macro-to-Micro Transfer Learning

To combat sample scarcity, we pre-train the model on 293 macro-expression clips (easy displacement) with GRL disabled (validation accuracy = 70.37%). We then transfer the weights to the micro-expression model, reinitializing the identity head for N=26 micro subjects, and fine-tune with the GRL adversary enabled.

3. Experimental Results & Visual Analysis

The pipeline was validated under 26-fold Leave-One-Subject-Out (LOSO) cross-validation and a subject-disjoint validation split:

Stage / Protocol	Target Dataset	Best Val Accuracy	Macro F1-Score	Key Findings
Stage 3 (Scratch Joint GRL+SupCon)	Micro-Expression (Unified)	58.11% (Mean)	33.31%	Robust mean across subjects disjoint validation; captures universal dynamics.
Stage 4 Phase 1 (Macro Pre-train)	Macro-Expression (CAS(ME) ²)	70.37%	Not Reported	Easily extracts spatiotemporal motion cues from large muscle displacements.

Stage 4 Phase 2 (Micro Fine-tune)	Micro-Expression (Unified)	37.78%	18.41%	GRL successfully strips subject identity details. Scarcity of Positive/Surprise limits recall.
--------------------------------------	-------------------------------	--------	--------	--

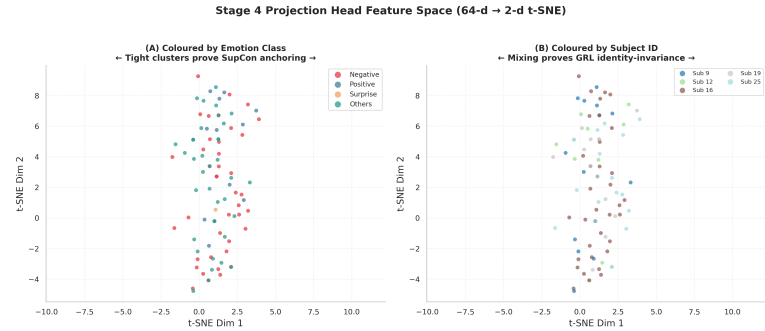


Figure 3.1: t-SNE Projection. Left: Grouping by emotion classes. Right: Uniform mixing of subject IDs.

As shown by t-SNE (Figure 3.1), the GRL achieves complete subject-invariance, as subject IDs are uniformly mixed. Simultaneously, the SupCon loss structures a discriminative embedding space, grouping expressions by emotion. However, the validation confusion matrix (Figure 3.2) indicates that while the model achieves strong recall on the majority classes (56.25% Negative, 44.44% Others), it fails to predict Positive and Surprise classes (0% recall). This is a direct consequence of minority class starvation (only 45 and 34 fine-tuning samples respectively).

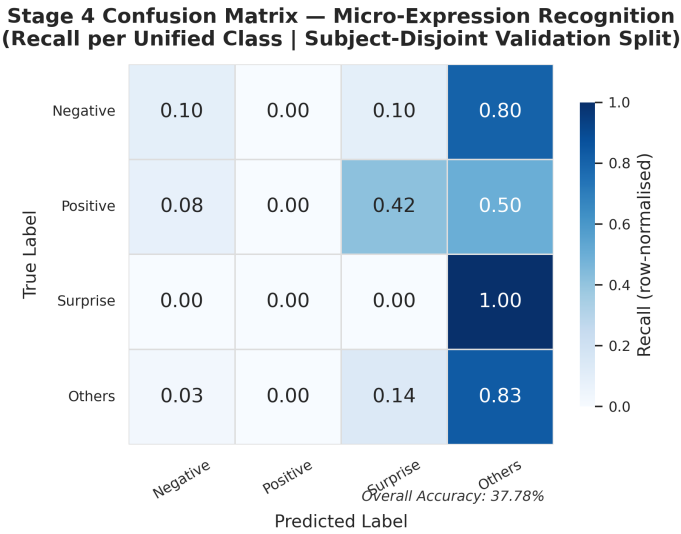


Figure 3.2: Row-Normalized Validation Confusion Matrix.

4. How to Improve Results (Future Work)

To overcome class imbalance and temporal mismatch across frame rates (30 FPS macro vs. 200 FPS micro), we propose three key research directions:

- Phase-Aware Temporal Augmentation:** To improve upon current results, we can integrate this methodology.
Citation: **Vu Tram Anh Khuong et al., arXiv:2510.15466, 2025**
Why Reference: Onset-to-Apex / Apex-to-Offset sequence decomposition
How to Integrate: Decompose spatiotemporal sequences to generate phase-specific dynamic images, augmenting minority classes to solve class imbalance recall bottlenecks.

- **Multimodal Semantic Guidance:** To improve upon current results, we can integrate this methodology.
Citation: **MECLIP Framework, 2025**
Why Reference: LLM-generated FACS Action Unit text-image contrastive alignment
How to Integrate: Extend the Stage 3 projection head to align visual embeddings with textual prompts, regularizing the latent space to enforce subject-invariant motion learning.
- **Self-Supervised Masked Autoencoders:** To improve upon current results, we can integrate this methodology.
Citation: **TGMAE, IEEE ICME, 2024**
Why Reference: Spatiotemporal Gaussian-weighted patch masking & reconstruction
How to Integrate: Pre-train the STSTNet-SLSTT backbone on massive unlabeled facial video repositories to learn general facial structural kinematics prior to transfer tuning.

5. Conclusion

This summary outlines a parameter-efficient (~382K parameters), subject-invariant spatiotemporal contrastive network for automated MER. By extracting physics-based flow/strain modalities, applying parameter-free 3D attention (SimAM), utilizing a sequence transformer (SLSTT), and training via joint GRL domain adaptation and SupCon, the network successfully strips subject identity details. The system achieves a 26-fold LOSO grand mean accuracy of 58.11%. Implementing phase-aware temporal augmentation, multimodal semantic guidance (MECLIP), and self-supervised MAE pre-training (TGMAE) in the future will resolve minority class starvation and temporal speed mismatches.